3B
COMPOUNDS & THE MOLE

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

MIOY T. HUYNH

LEARNING OBJECTIVES

CHAPTER 3

1. Interpret chemical formulas.
2. Write chemical formulas to represent simple inorganic chemical compounds.
3. Name simple inorganic chemical compounds.
4. Calculate the number of moles from the number of particles and vice versa.
5. Calculate the mass of 1 mole of a substance based on its chemical formula and atomic masses from the periodic table.
6. Calculate the percent composition by mass of a compound based on its chemical formula and atomic masses from the periodic table.
7. Determine the empirical and molecular formulas of a compound from percent composition or other mass-ratio data.
8. Determine the empirical formula for a compound based on data from a combustion analysis.

Terminology

Chemistry in Context

It's important to understand chemical vocabulary.

Atom: Smallest part of an element that is still that element.

Molecule: Two or more atoms joined together and acting as a unit.

Compound: Two or more *different* elements joined together.

Chemists/Scientists use the term **element** in different ways though.

Chemical Formulas

3.1

Atomic elements: Most elements exist in nature as individual neutral atoms.

Molecular elements: A few elements exist in nature as small molecules containing two or more atoms bonded together.

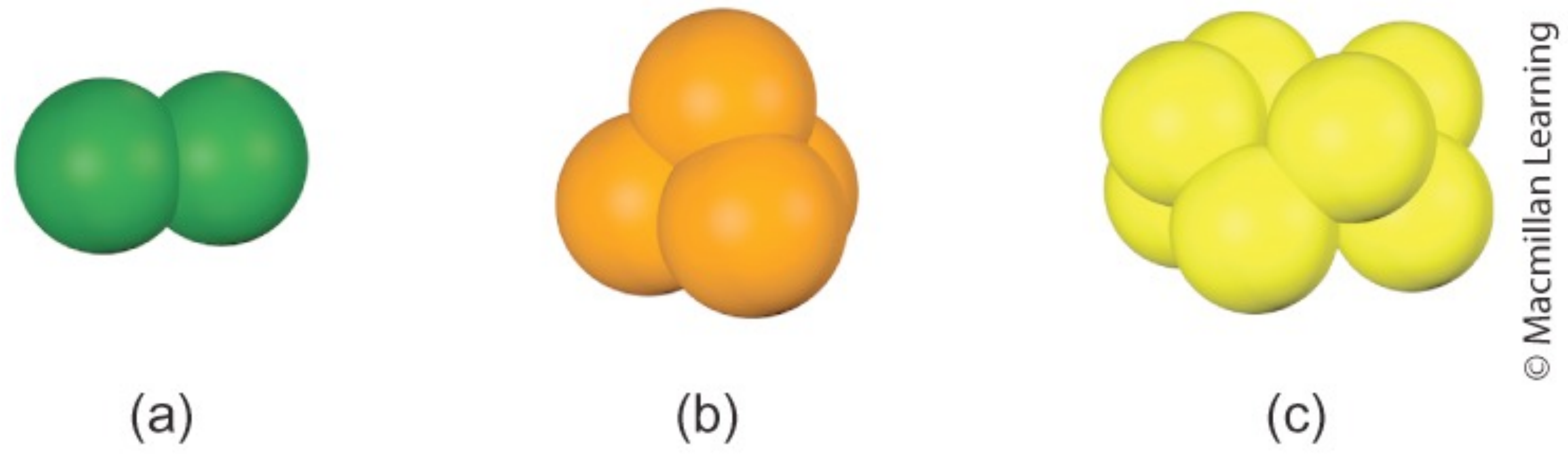


FIGURE 3.2 Examples of molecular elements (a) Cl_2, (b) P_4, (c) S_8.

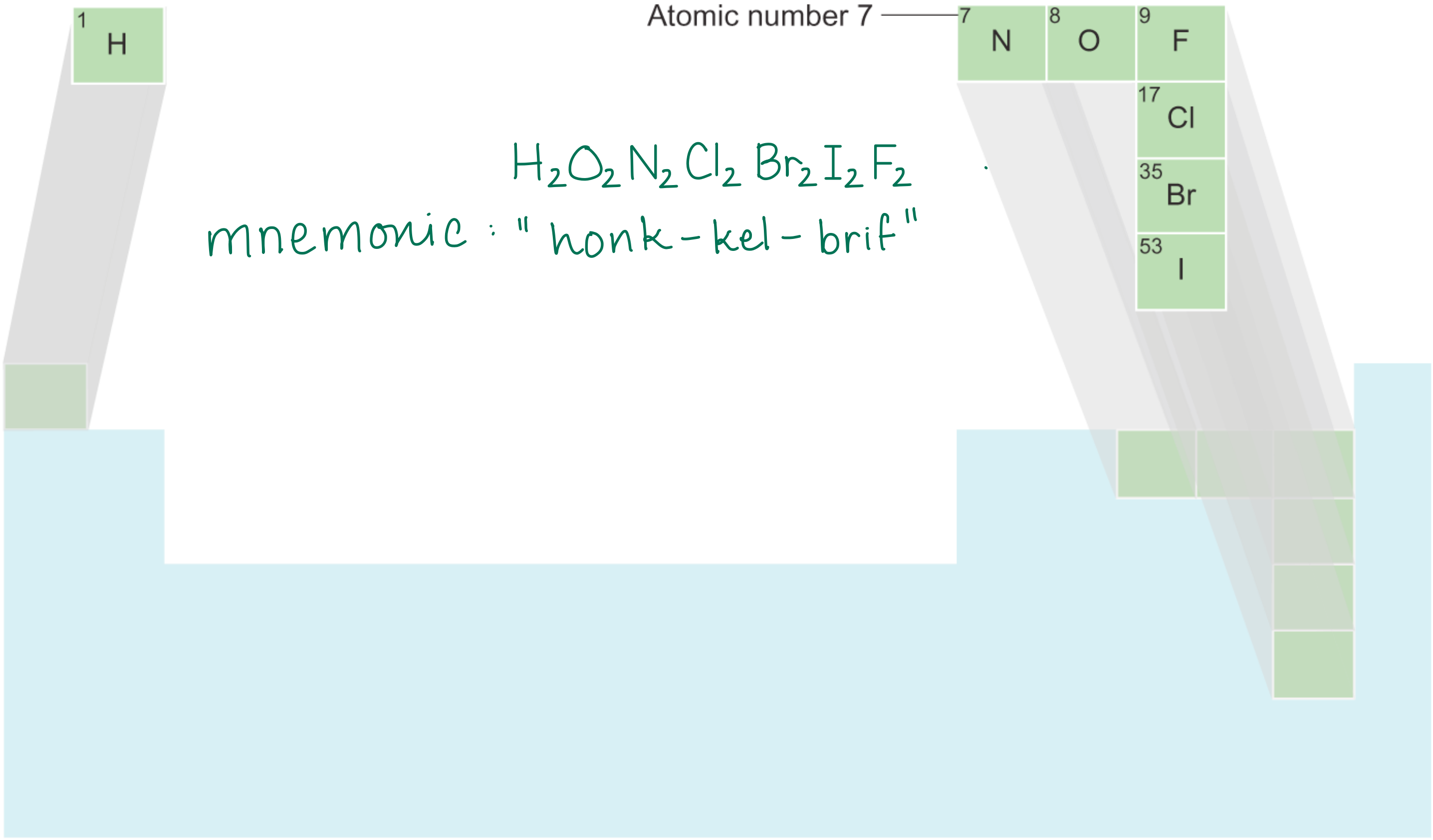
Chemical Formulas

3.1

Molecular diatomic elements: Seven elements occur as diatomic molecules when they are in their elemental/natural form.

- Hydrogen $\longrightarrow H_2$
- Nitrogen $\longrightarrow N_2$
- Oxygen $\longrightarrow O_2$
- Fluorine $\longrightarrow F_2$
- Chlorine $\longrightarrow Cl_2$
- Bromine $\longrightarrow Br_2$
- Iodine $\longrightarrow I_2$

tip get in habit of translating these 7 into diatomics



$H_2O_2N_2Cl_2Br_2I_2F_2$

mnemonic : "honk-kel-brif"

FIGURE 3.3 The Seven Elements that Exist as Diatomic Molecules

Chemical Formulas

3.1

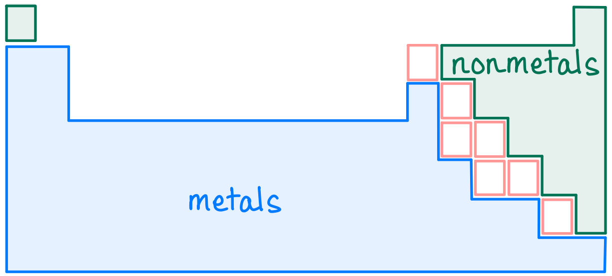
Compounds can exist in different forms.

- **Molecular or covalent compounds**: Exist as individual particles (called molecules) made up of bonded nonmetal elements.

nonmetals only

- **Ionic compounds**: Exist not as individual particles but as extended, 3D lattice structures. The simplest repeating unit is called a formula unit.

metals + nonmetals



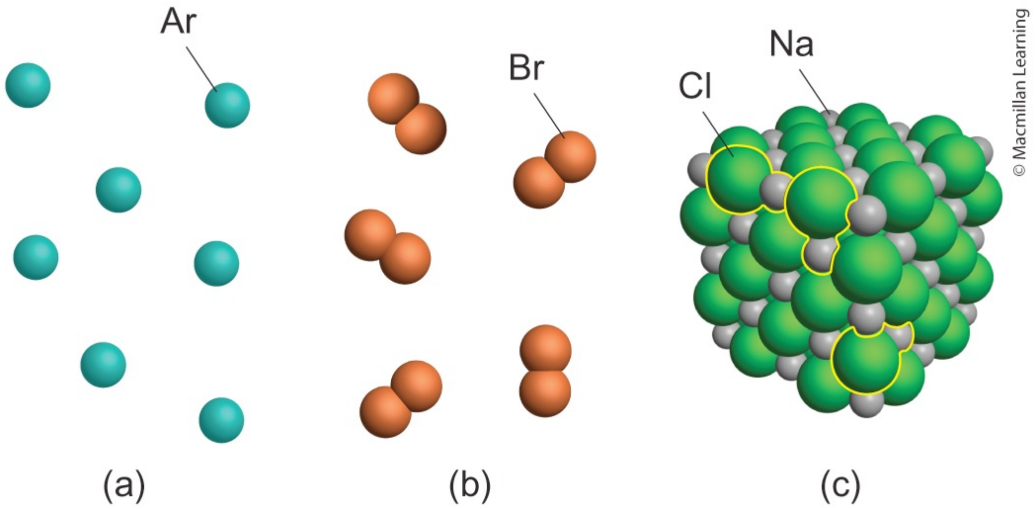


FIGURE 3.1 (a) atoms of Ar, (b) molecules of Br_2, (c) lattice of NaCl

Chemical Formulas

3.1

Chemical formulas combine elemental symbols and subscripts to represent chemical compounds.

- Provide information about the relative number of atoms of each element
- Sometimes also give information about the arrangement of atoms in the molecule.

very generic term

The group of atoms represented by a chemical formula is called a **formula unit**.

- An atom or molecule of an uncombined element, a molecule of a covalent compound, or a set of ions in an ionic compound

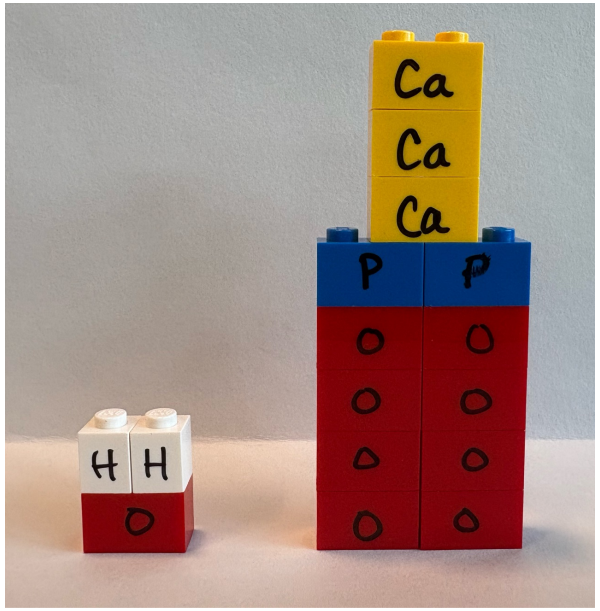
examples of formula units:	Na, Li, Ar	atomic elements
	H_2, Br_2, Cl_2	molecular elements
	H_2O, CH_4	molecular/covalent compounds
	NaCl, $Ca_3(PO_4)_2$	ionic compounds

Chemical Formulas

3.1

Chemists combine chemical symbols with subscripted numbers to form chemical formulas.

The subscripted numbers indicate the number of atoms of each element present.



$$H_2O$$

$$Ca_3(PO_4)_2$$

1 formula unit $Ca_3(PO_4)_2$: 3 atoms Ca : 2 atoms P : 8 atoms O

Naming Binary Covalent Compounds

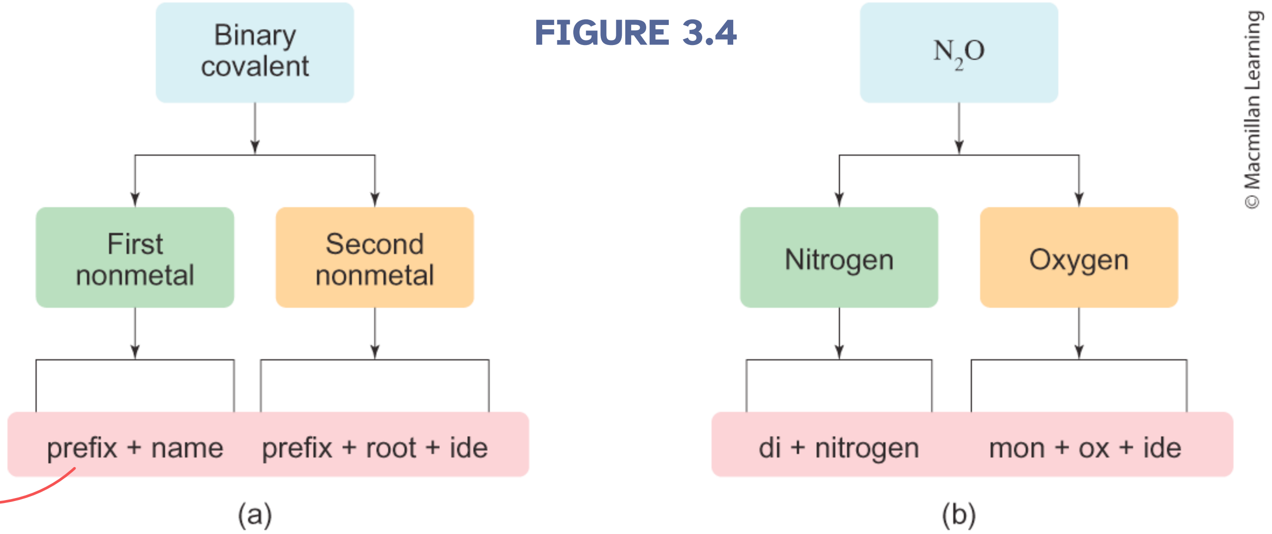
3.2

Molecular (or covalent) compounds form between two or more nonmetals.

To name a binary molecular compound:

2 elements

FIGURE 3.4



indicates # of atoms

- *Note 1: If the element name starts with an "o" then drop the last "o" or the last "a" of the prefix.*
- *Note 2: Do not use the prefix "mono" for the first nonmetal element.*

Number of atoms	Prefix
1	mono-
2	di-
3	tri-
4	tetra-
5	penta-
6	hexa-
7	hepta-
8	octa-
9	nona-
10	deca-

Formulas for Ionic Compounds

3.3

Ions are single atoms (***monatomic ions***) or groups of atoms (***polyatomic ions***) that have gained or lost electrons and therefore have a net negative or a net positive charge.

- Cations: positively charged ions, less electrons than protons, usually metals
- Anions: negatively charged ions, more electrons than protons, usually nonmetals

You should be familiar with the stable, **monatomic ions** of the orange and green elements.

1 1A	2 2A	3	4	5	6	7	8	9	10	11	12	13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
H +1																	
Li +1	**Be** +2													**N** −3	**O** −2	**F** −1	
Na +1	**Mg** +2											**Al** +3		**P** −3	**S** −2	**Cl** −1	
K +1	**Ca** +2		**Ti** *variable*	**V** *variable*	**Cr** *variable*	**Mn** *variable*	**Fe** *variable*	**Co** *variable*	**Ni** *variable*	**Cu** *variable*	**Zn** +2				**Se** −2	**Br** −1	
Rb +1	**Sr** +2								**Pd** *variable*	**Ag** +1	**Cd** +2		**Sn** *variable*			**I** −1	
Cs +1	**Ba** +2								**Pt** *variable*	**Au** *variable*	**Hg** *variable*		**Pb** *variable*				
Fr +1	**Ra** +2																

FIGURE 3.7

Formulas for Ionic Compounds

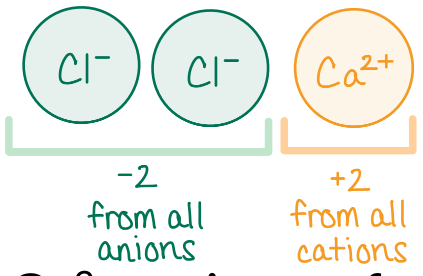
3.3

<u>Binary ionic compounds</u> form when metal atoms combine with nonmetal atoms.

- Metal atoms give one or more electrons (forming cations) to nonmetal atoms (which form anions).

Ionic compounds are electrically neutral overall

- Total positive charge of cations = Total negative charge of anions
- Ca forms only the Ca^{2+} ion
- Cl forms only the Cl^- ion



- Two Cl^- anions are required for every one Ca^{2+} cation to form the neutral ionic compound: $CaCl_2$

Formulas for Ionic Compounds

3.3

<u>Polyatomic ions</u> are *groups* of two or more bonded atoms that have lost or gained electrons and thus have an overall net charge.

You should be familiar with the following common polyatomic ions.

Ion Name	Formula	Ion Name	Formula
Ammonium	NH_4^+	Permanganate	MnO_4^-
Hydroxide	OH^-	Perchlorate	ClO_4^-
Cyanide	CN^-	Chlorate	ClO_3^-
Acetate	$C_2H_3O_2^-$ or CH_3COO^-	Chlorite	ClO_2^-
Nitrite / Nitrate	NO_2^- / NO_3^-	Hypochlorite	ClO^-
Sulfite / Nitrate	SO_3^{2-} / SO_4^{2-}	Perbromate	BrO_4^-
Phosphite / Phosphate	PO_3^{3-} / PO_4^{3-}	Bromate	BrO_3^-
Carbonate	CO_3^{2-}	Bromite	BrO_2^-
Hydrogen sulfite / sulfate	HSO_3^- / HSO_4^-	Hypobromite	BrO^-
Hydrogen phosphite / phosphate	HPO_3^{2-} / HPO_4^{2-}	Periodate	IO_4^-
Hydrogen carbonate (bicarbonate)	HCO_3^{2-}	Iodate	IO_3^-
Chromate	CrO_4^{2-}	Iodite	IO_2^-
Dichromate	$Cr_2O_7^{2-}$	Hypoiodite	IO^-

TABLE 3.4

per- ☐ -ate even more O atoms

☐ -ate more O atoms

☐ -ite less O atoms

hypo- ☐ -ite even less O atoms

☐ -ide no O atoms

Naming Ionic Compounds

3.4

To name an ionic compound:

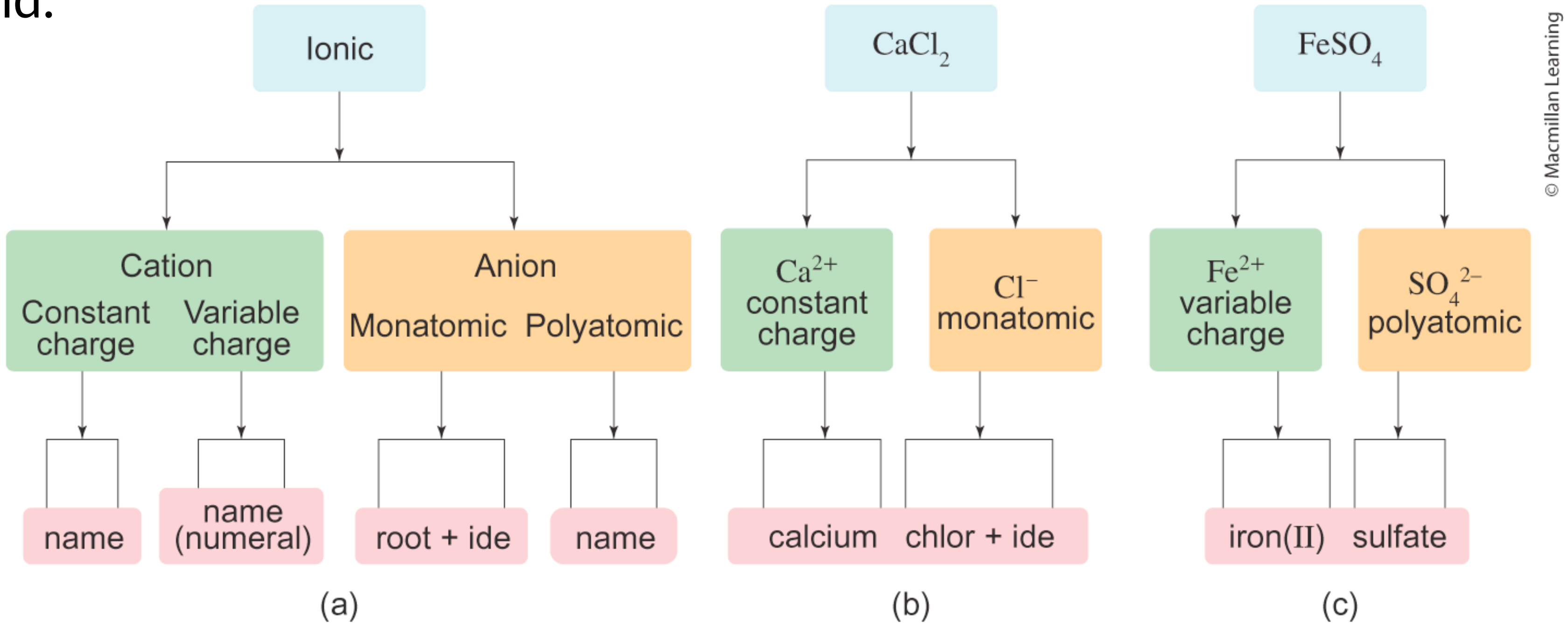


FIGURE 3.10

1. *Do not use prefixes.*
2. *If the metal is a group 1 (alkali) metal, a group 2 (alkaline earth) metal, Al, Zn, or Ag do not use Roman numerals.*
3. *If the metal is a transition metal, use Roman numerals because the charge is variable. Determine charge from anion.*
4. *If there is a polyatomic ion, do not modify/change the name of the polyatomic ion.*

IN-CLASS QUESTION 1

Learning Objective(s): 3.2

TIP use anion charges to get metal cation charges

Name the chemical compounds:

TIP don't modify polyatomics!

formula →	compound type →	ions present →	name
$Ca(MnO_4)_2$	ionic	Ca^{2+} MnO_4^- MnO_4^-	calcium permanganate
SF_6	covalent	n/a	sulfur hexafluoride
$Cu(CN)_2$	ionic	Cu^{2+} CN^- CN^-	copper(II) cyanide
Ag_2CrO_4	ionic	Ag^+ Ag^+ CrO_4^{2-}	silver chromate
P_4O_{10}	covalent	n/a	tetraphosphorus decoxide
$Co_3(PO_4)_2$	ionic	Co^{2+} Co^{2+} Co^{2+} PO_4^{3-} PO_4^{3-}	cobalt(II) phosphate

TIP no prefixes

TIP fluoride not flouride

TIP use prefixes

TIP for transition metal Roman numeral is charge except Ag^+, Cd^{2+}, Zn^{2+}

TIP drop first vowel to avoid "ao" or "oo"

IN-CLASS QUESTION 2

Learning Objective(s): 3.2

from chart of polyatomics SO_3^{2-}

How many sulfite ions are required to form a neutral ionic compound with the ion formed when vanadium has 20 electrons? Then name the compound.

from PT: vanadium has $Z=23$, so neutral V has 23 e^-,
cation V^+ has 22 e^-,
cation V^{2+} has 21 e^-,
cation V^{3+} has 20 e^-

Construct formula through charge balance

$$\underbrace{V^{3+} \quad V^{3+}}_{+6} \quad \underbrace{SO_3^{2-} \quad SO_3^{2-} \quad SO_3^{2-}}_{-6} \longrightarrow \underbrace{V_2(SO_3)_3}_{\emptyset}$$

from all cations from all anions neutral

Convert formula to name

vanadium(III) sulfite

don't modify polyatomics!

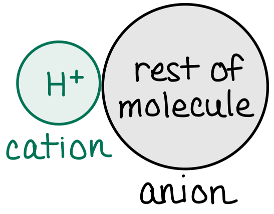
transition metal: Roman numeral is charge

Naming Acids

3.5

Acids are a special class of hydrogen-containing covalent compounds that, when dissolved in water:

- Release one or more H^+ ions (called ionizable hydrogen atoms)
- Form an anion with a −1 charge for each H^+ ion that was released.



Typically, the ionizable H atom is written at the beginning of the chemical formula to denote it is an acid.

H_2CO_3 vs. CH_2O

acid not acid

There are two types of acids:

- **Binary acids**: contain H atoms and atoms of an element from group 6A/16 or 7A/17.
  e.g. HCl, HF, H_2S
- **Oxyacids**: contain H atoms bonded to oxyanions (polyatomic ions containing oxygen).
  e.g. $HClO_4$, H_2SO_4

Naming Acids

3.5

To name an acid compound, first determine whether the acid is a binary acid (left diagrams) or an oxyacid (right diagrams).

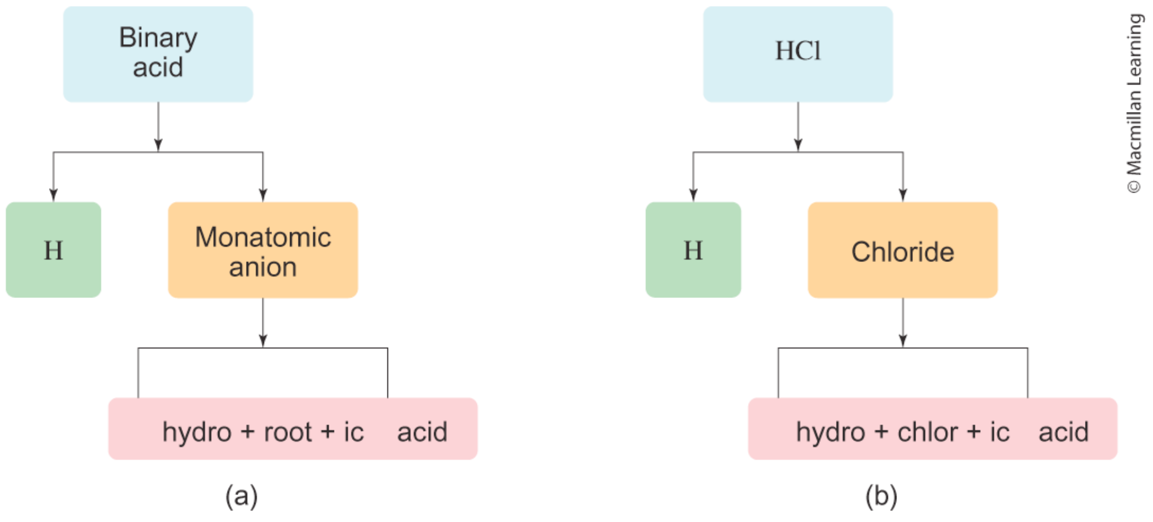


FIGURE 3.13

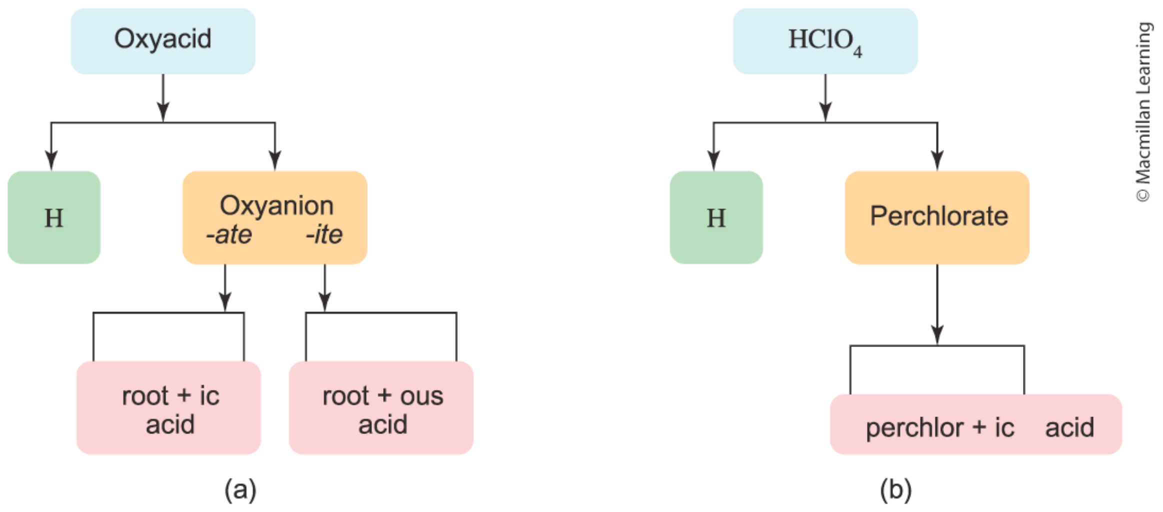


FIGURE 3.14

EXAMPLE PROBLEM 1

Learning Objective(s): 3.3

Name the following acids:

HCl binary acid

$HCl \rightarrow H^+ + Cl^-$

hydrochloric acid

chloride

add "hydro–" prefix; change -ide to -ic acid

HNO_2 oxyacid

$HNO_2 \rightarrow H^+ + NO_2^-$

nitrous acid

nitrite

change -ite to -ous acid

$HBrO_4$ oxyacid

$HBrO_4 \rightarrow H^+ + BrO_4^-$

perbromic acid

perbromate

change -ate to -ic acid

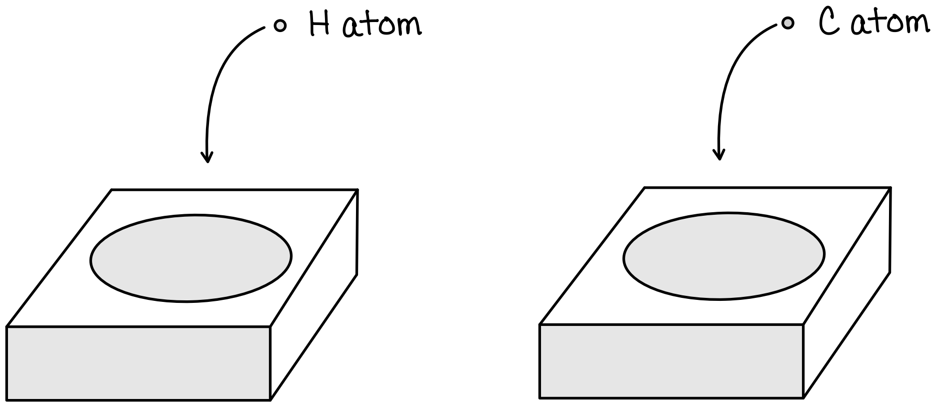
The Mole

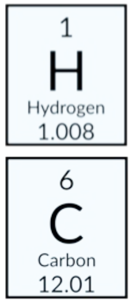
3.7

A **mole** is a means of counting the large number of particles in samples.

- One mole was defined as the number of atoms in exactly 12 grams of ^{12}C (carbon-12).

- 1 mole contains an **Avogadro's number** **(N_A = 6.022 × 10^{23})** of particles.





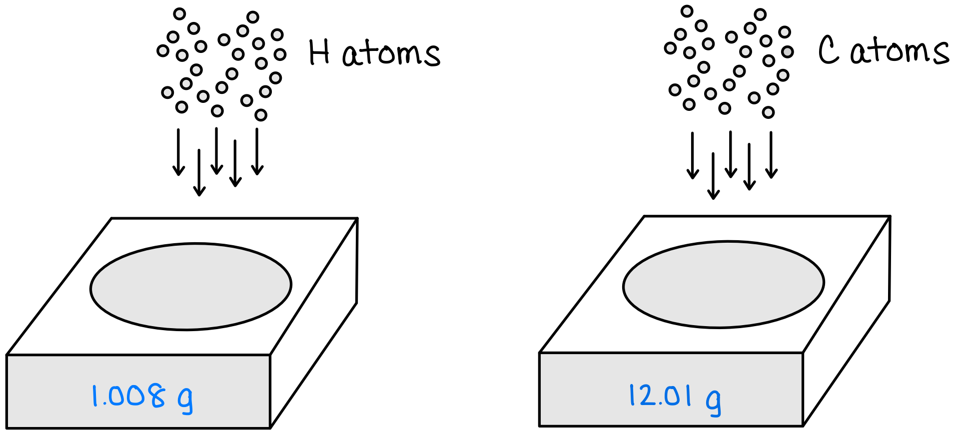
	1 atom	
H	1.008 amu	
C	12.01 amu	

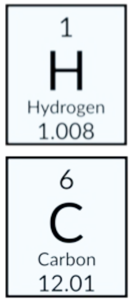
The Mole

3.7

A **mole** is a means of counting the large number of particles in samples.

- One mole was defined as the number of atoms in exactly 12 grams of ^{12}C (carbon-12).
- 1 mole contains an **Avogadro's number** ($N_A = 6.022 \times 10^{23}$) of particles.





	1 atom	N_A atoms
H	1.008 amu	1.008 g
C	12.01 amu	12.01 g

Molar Mass

3.8

Atomic mass: For an element, the atomic mass is the average mass of one atom, expressed in atomic mass units (amu).

- One atomic mass unit, amu, is equal to 1 gram (g) divided by Avogadro's number such that: **$1 \text{ amu} = 1.661 \times 10^{-24}$ g**

Molar mass: For an element, the molar mass is the mass in grams of exactly 1 mole (or 6.022×10^{23} atoms) of that element

- Molar mass allows conversion between mass and the number of moles.

$$\frac{6.022 \times 10^{23} \text{ "things"}}{1 \text{ mol "things"}} \text{ or } \frac{1 \text{ mol "things"}}{6.022 \times 10^{23} \text{ "things"}} \qquad \frac{12.01 \text{ g C}}{1 \text{ mol C}} \text{ or } \frac{1 \text{ mol C}}{12.01 \text{ g C}}$$

$$\text{number of "things"} \xleftrightarrow[\text{Avogadro's \#}]{} \text{moles} \xleftrightarrow[\text{molar mass}]{} \text{grams}$$

"things" = atoms, molecules, formula units, etc.

Molar Mass

3.8

The molar mass of a compound is the sum of the molar masses of all the atoms in a formula unit of the compound.

ratios don't change!

1.008 amu + 1.008 amu + 15.999 amu

1 molecule H_2O : 2 atoms H : 1 atoms O : 18.015 amu

2 × [1 molecule H_2O : 2 atoms H : 1 atoms O : 18.015 amu]

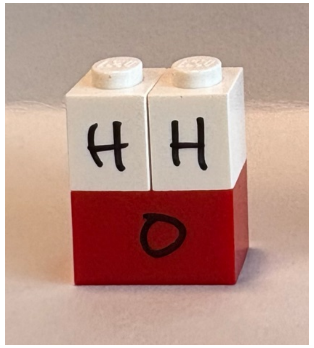
3 × [1 molecule H_2O : 2 atoms H : 1 atoms O : 18.015 amu]

N_A × [1 molecule H_2O : 2 atoms H : 1 atoms O : 18.015 amu]

1 mol "x" = N_A "x"

1.008 g + 1.008 g + 15.999 g

1 mole H_2O : 2 mole H : 1 mole O : 18.015 g



EXAMPLE PROBLEM 2

Learning Objective(s): 3.4

molar mass = 227.132 g/mol

mass

How many moles of $C_7H_5N_3O_6$ are in 300. g of $C_7H_5N_3O_6$?

How many molecules of $C_7H_5N_3O_6$ are in 300. g of $C_7H_5N_3O_6$?

$$\underbrace{300.\text{ g } C_7H_5N_3O_6}_{\text{grams}} \times \underbrace{\frac{1\text{ mol } C_7H_5N_3O_6}{227.132\text{ g } C_7H_5N_3O_6}}_{\text{molar mass}} = \underbrace{1.32_1\text{ mol } C_7H_5N_3O_6}_{\text{moles}} \times \underbrace{\frac{6.022\times10^{23}\text{ molecules } C_7H_5N_3O_6}{1\text{ mol } C_7H_5N_3O_6}}_{\text{Avogadro's \#}} = \underbrace{7.95\times10^{23}\text{ molecules } C_7H_5N_3O_6}_{\text{number of "things"}}$$

General Strategy:

$$\text{number of "things"} \xleftrightarrow[\text{Avogadro's \#}]{\frac{6.022\times10^{23}\text{ "things"}}{1\text{ mol "things"}} \text{ or } \frac{1\text{ mol "things"}}{6.022\times10^{23}\text{ "things"}}} \text{moles} \xleftrightarrow[\text{molar mass}]{\frac{12.01\text{ g C}}{1\text{ mol C}} \text{ or } \frac{1\text{ mol C}}{12.01\text{ g C}}} \text{grams}$$

"things" = atoms, molecules, formula units, etc.

EXAMPLE PROBLEM 4

Learning Objective(s): 3.4, 3.5

transition metal : Roman numeral is charge

from chart of polyatomics

Fe^{3+} CO_3^{2-}

How many moles of carbon are present in 125.0 g of iron(III) carbonate?

ionic compound

Convert name to formula.

$$\underbrace{Fe^{3+}\ Fe^{3+}}_{\substack{+6 \\ \text{from all cations}}}\ \underbrace{CO_3^{2-}\ CO_3^{2-}\ CO_3^{2-}}_{\substack{-6 \\ \text{from all anions}}} \longrightarrow \underbrace{Fe_2(CO_3)_3}_{\substack{\varnothing \\ \text{neutral}}}$$

molar mass = 291.714 g/mol

General Strategy: $\text{mass } Fe_2(CO_3)_3 \xrightarrow{\text{molar mass}} \text{mol } Fe_2(CO_3)_3 \xrightarrow{\text{chemical formula ratio}} \text{mol C}$

1 formula unit $Fe_2(CO_3)_3$: 2 atoms Fe : 3 atoms C : 9 atoms O

1 mol $Fe_2(CO_3)_3$: 2 mol Fe : 3 mol C : 9 mol O

$$125.0 \text{ g } Fe_2(CO_3)_3 \times \frac{1 \text{ mol } Fe_2(CO_3)_3}{291.714 \text{ g } Fe_2(CO_3)_3} = 0.4285 \text{ mol } Fe_2(CO_3)_3 \times \frac{3 \text{ mol C}}{1 \text{ mol } Fe_2(CO_3)_3} = 1.285 \text{ mol C}$$

$$125.0 \text{ g } Fe_2(CO_3)_3 \times \frac{1 \text{ mol } Fe_2(CO_3)_3}{291.714 \text{ g } Fe_2(CO_3)_3} \times \frac{3 \text{ mol C}}{1 \text{ mol } Fe_2(CO_3)_3} = 1.285 \text{ mol C}$$

as one big dimensional analysis conversion

IN-CLASS QUESTION 3

Learning Objective(s): 3.4

Determine the number of oxygen atoms in a sample of 2.67 moles of hydrogen peroxide (H_2O_2)?

1 molecule H_2O_2 : 2 atoms H : 2 atoms O

1 mol H_2O_2 : 2 mol H : 2 mol O

ratios of atom-to-atom or mol-to-mol are equivalent

General Strategy: mol H_2O_2 $\xrightarrow{\text{chemical formula ratio}}$ mol O $\xrightarrow{\text{Avogadro's \#}}$ atoms O

$$2.67 \text{ mol } H_2O_2 \times \frac{2 \text{ mol O}}{1 \text{ mol } H_2O} = 5.34 \text{ mol O} \times \frac{6.022\times10^{23} \text{ atoms O}}{1 \text{ mol O}} = 3.22\times10^{24} \text{ atoms O}$$

Alternatively: mol H_2O_2 $\xrightarrow{\text{Avogadro's \#}}$ molecules H_2O_2 $\xrightarrow{\text{chemical formula ratio}}$ atoms O

$$2.67 \text{ mol } H_2O_2 \times \frac{6.022\times10^{23} \text{ molecules } H_2O_2}{1 \text{ mol } H_2O_2} = 1.60_8\times10^{24} \text{ molecules } H_2O_2 \times \frac{2 \text{ atoms O}}{1 \text{ molecule } H_2O_2} = 3.22\times10^{24} \text{ atoms O}$$

A Note on Chemists' Vocabulary

Chemistry in Context

An atom is a single particle that represents an element.

A molecule is a group of atoms that are bonded together.

A compound is a group of atoms of two or more *different* elements.

A formula unit is an extremely generic term that can refer to single atoms, ions, molecules, or ionic compounds.

- Likewise, formula weight or formula mass are generic terms that may refer to atoms, ions, molecules, or ionic compounds.
- Though infrequently used, this term is best reserved for ionic compounds since they do not exist as individual molecules.

Atomic weight or atomic mass refers to atoms.

Molecular weight or molecular mass refers to molecules.

Molar mass can be used to refer to the mass of one mole of atoms, molecules, or formula units of ionic compounds.

Few chemists actually use the terms formula weight or formula mass and use molar mass and molecular weight somewhat interchangeably.